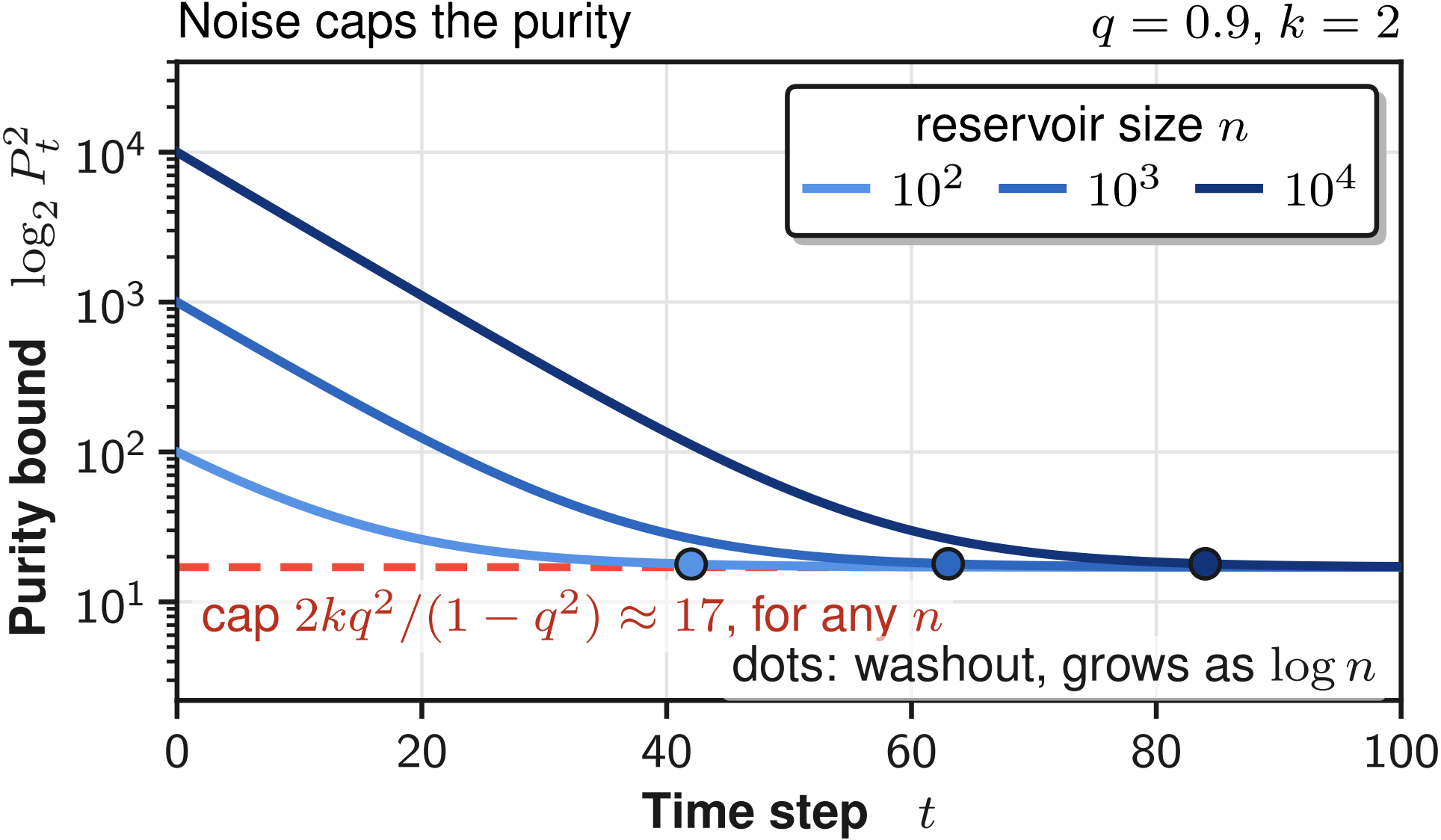

Noise caps the purity
$q = 0.9$, $k = 2$
reservoir size n
10^2
10^3
10^4
Purity bound $\log_2 P_t^2$
10^4
10^3
10^2
10^1
cap $2kq^2/(1-q^2) \approx 17$, for any n
dots: washout, grows as $\log n$
0
20
40
60
80
100
Time step t